

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-	Team No-	Project Title- Chat Application with End-to-End Encryption
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1. Abstract

In today's digital communication landscape, most messaging platforms route user data through centralized servers that can access message content, raising serious privacy concerns. This project proposes a Chat Application with End-to-End Encryption that guarantees private one-to-one and group messaging where the server never sees plaintext content. The system uses a Signal-style key exchange protocol built on libsodium to establish secure session keys between users, ensuring only intended recipients can decrypt messages. The application is built using React.js for the frontend, Node.js and Socket.io for real-time backend communication, and MongoDB for storing encrypted message metadata and user information.

Core features include secure user authentication with automatic key generation, end-to-end key exchange for establishing trust between users, fully encrypted one-to-one and group messaging, encrypted media sharing for images and files, and encrypted message synchronization and backup across devices. By combining modern web technologies with proven cryptographic protocols, the system addresses the growing need for privacy-preserving communication tools while maintaining a responsive, user-friendly experience. The proposed solution demonstrates that strong security and usability can coexist, offering a practical model for building trustworthy, privacy-first messaging platforms for personal and organizational use.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090042	Chakrika Panakala	
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3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

React.js, Node.js, Socket.io, MongoDB, Signal-style Key Exchange (libsodium)

Key Features: User Auth & Key Generation; E2E Key Exchange; Encrypted Messaging; Group Chats; Media Sharing (Encrypted); Message Sync & Backup

Guide Name and Signature (remarks if any) _____